Requirement Document: Smart Content Summarizer (Topic 1)

1. Document Overview

This document defines the requirements, workflow, inputs, outputs, quality expectations, and deliverables for Smart Content Summarizer built using Google Opal.

2. Objective and Purpose

Purpose Statement: Enable learners to build and configure workflows within Google Opal that orchestrate AI reasoning, tools, and output integrations.

Primary Tasks

1. Authenticate and access the Google Opal visual editor.
2. Create a workflow containing user input, generation, and output steps.
3. Integrate a webpage retrieval tool to fetch article content.
4. Configure the Generate Content Summary step with Objective, Output Format, and multiple context variables.
5. Build a Markdown output page and preview the final application.

3. Scope

In Scope

- Collecting summarization instructions.
- Collecting an article URL.
- Retrieving webpage text using the connected Get Webpage tool.
- Passing multiple context variables to the summary step.
- Generating a summary based on retrieved content.
- Rendering the source URL, instructions, summary, and original content in Markdown.
- Testing the workflow in Preview.

Out of Scope

- Custom SQL database integration.
- Guaranteed retrieval of every webpage.

- Guaranteed factual accuracy without human review.
- Raw HTML in output nodes where Markdown is expected.
- Payment, account-management, or unrelated external integrations.

4. Functional Requirements

FR-01: Collect User Instructions

The application must collect the user's preferred summary length, format, language, tone, and level of detail.

FR-02: Collect Article URL

The application must request the URL of the webpage to summarize.

FR-03: Retrieve Web Content

The Fetch content step must pass the supplied URL to the connected Get Webpage tool and return retrieved text.

FR-04: Generate Summary

The Generate Content Summary step must use both the retrieved article and user instructions.

FR-05: Follow Formatting Requirements

The summary must follow the user's requested format where feasible. For example, a request for exactly three bullet points should produce three bullet points.

FR-06: Handle Retrieval Errors

If the webpage cannot be retrieved or the retrieved content is empty, the application must report the failure instead of inventing a summary.

FR-07: Render Final Output

The final page must display:

- User instructions.
- Source URL as a clickable Markdown link.
- Generated summary.

- Original retrieved webpage content.

5. Input Specification

Input 1: User Instructions

Prompt: "Please provide your instructions for summarizing the article. You can specify the desired length, format, language, tone, level of detail, or any other requirements."

Input 2: Get Content Source

Prompt: "Please provide the URL of the article you would like me to summarize."

6. Output Specification

A rendered Markdown webpage with the title **Smart Content Summarizer** and sections for User Instructions, Source Article, Executive Summary, and Original Web Content.

7. Quality Requirements

- **Readability:** Use clear headings and consistent formatting.
- **Traceability:** Show the original source URL.
- **Grounding:** Base the summary on retrieved article content.
- **Instruction adherence:** Respect the user's requested format and tone.
- **Error handling:** Clearly report retrieval failures.
- **Usability:** Keep the main summary easy to locate.

8. Deliverables

1. A working Google Opal workflow.
2. A public GitHub repository containing the `prompts/` folder and required Markdown guides.
3. A project explanation video showing the workflow, configuration, testing, and final output.

9. Acceptance Criteria

The project is considered ready for demonstration when:

- Both inputs can be submitted.
- The retrieval step attempts to fetch the supplied URL.
- The summary step receives both article content and instructions.
- The final page displays the source, instructions, summary, and original content.
- The example test succeeds.
- Retrieval errors are handled without fabricated summaries.
- The required repository files and demonstration video are prepared.

10. Known Limitations

Webpage retrieval depends on the connected tool and the accessibility of the target webpage. Successful retrieval of one URL does not guarantee successful retrieval of every URL. Generated summaries should be checked against their sources.